

CNG KSP Feature Matrix

Every capability observed across shipping CNG Key Storage Providers, with this project's coverage and — where a gap exists — what closing it would take. Generated 2026-09-23.

101	65	12	24	64%	25
Capabilities tracked	Covered	Partial	Not covered	Fully covered	Actionable gaps

How to read this

Status	Meaning	Ev	Evidence for the market claim
Covered	Implemented and exercised by the test suite.	A	Read from source code or an authoritative API reference that enumerates the provider's own algorithm list.
Partial	Implemented, but reachable only through a non-standard identifier or with a stated caveat.	B	Stated in vendor or Microsoft documentation surfaced through search; primary page not read in full.
Not covered	Absent. The Gap column says what closing it would take.	C	Drawn from HSM datasheets describing the device rather than its KSP. Treat as a lead, not a fact.

Effort on gaps: S = a few lines to a day · M = days · L = a substantial piece of work. **N/A** marks a deliberate position or an external blocker rather than a backlog item — those are excluded from the actionable-gap count.

ID	Feature	Detail	Market support	Ev	Status	Gap — what closing it would take	Eff
PROVIDER INTERFACE							
IFACE-01	GetKeyStorageInterface export	Sole exported symbol; returns the KSP function table	All providers	A	Covered	—	—
IFACE-02	OpenProvider / FreeProvider	Provider handle lifecycle	All providers	A	Covered	—	—
IFACE-03	GetProviderProperty	Name / Version / ImplType	All providers	A	Covered	—	—
IFACE-04	SetProviderProperty	Provider-level configuration	Vendor-specific	C	Partial	NCRYPT_PIN_PROPERTY is implemented: the PIN is handed to the session layer, and because sessions open lazily a PIN set before the first cryptographic call is the one used to log in. Token selection is read-only through this call by design — the session pool is already bound to a slot by then and PKCS#11 cannot move a session between tokens, so accepting the value would be worse than refusing it. Choose the token with SOFTHSM2_SLOT or SOFTHSM2_TOKEN_LABEL and read it back with KSP_SLOT_PROPERTY.	M
IFACE-05	NotifyChangeKey	Key-change notification handle	Microsoft KSP	B	Not covered	Returns ERROR_SUCCESS as a no-op. Real support needs a registry/file watcher on the token store and an event object per caller.	L
IFACE-06	GetOperationProperty	Async / overlapped operation state	Microsoft KSP	B	Not covered	Returns NTE_NOT_SUPPORTED. Only needed for overlapped I/O; PKCS#11 is synchronous so there is nothing to report.	L
IFACE-07	PromptUser	Interactive PIN / consent dialog	Smart card KSPs	B	Not covered	Returns NTE_NOT_SUPPORTED. Would need a UI thread and a window handle from NCRYPT_WINDOW_HANDLE_PROPERTY.	L
IFACE-08	FreeBuffer / FreeObject	Provider-allocated memory release	All providers	A	Covered	—	—
TABLE-01	Function table layout matches the SDK	NCRYPT_KEY_STORAGE_FUNCTION_TABLE slot order and membership	All providers	A	Partial	The table uses designated initialisers, so the SDK header assigns slots by name and a wrong name is a build error rather than a displaced slot. IsAlgSupported, EnumAlgorithms and VerifySignature implemented and wired; the invented GetOperationProperty and FreeObject slots removed. test_function_table.c covers ksp_main.c for the first time. Cannot be confirmed against the real ncrypt_provider.h until that header is available — see BUILD-01.	S

ID	Feature	Detail	Market support	Ev	Status	Gap — what closing it would take	Eff
BUILD-01	Builds for Windows at all	The shipped artefact is a Windows DLL	All providers	A	Partial	Phase 0 fixed all six root causes. MSVC compiles nine of the ten source files clean at /W3 /WX and links test_p11_layer. ksp_main.c cannot be built in hosted CI: ncrypt_provider.h does not exist on a windows-latest runner, established by a dir /s /b across C: and D: that returned only this repo's own test mock, after a WDK install that reported success. Needs a machine that has the header.	M
BUILD-02	Continuous integration	Automated build and test on the target platform	All providers	A	Covered	—	—
ASYMMETRIC RSA							
RSA-01	RSA key generation	2048 / 3072 / 4096 bits	All providers	A	Covered	—	—
RSA-02	RSA 1024 and below	512-1024 bits	MS Software KSP; Utimaco	B	Partial	Build-time opt-in only. KSP_RSA_MIN_BITS defaults to 2048; a legacy build can lower it, but the shipped default still rejects <2048 with NTE_BAD_LEN. Deliberate: silently allowing 1024 would weaken every deployment.	S
RSA-03	RSA above 4096	8192 / 16384 bits	MS Software KSP; Utimaco	B	Covered	—	—
RSA-04	RSA arbitrary size increments	64-bit (MS) or 8-bit (Utimaco) steps	MS Software KSP; Utimaco	B	Covered	—	—
RSA-05	Custom public exponent	Exponent other than 65537	Vendor-specific	C	Covered	—	—
RSA-06	PKCS#1 v1.5 signing	SHA-1 / 256 / 384 / 512	All providers	A	Covered	—	—
RSA-07	PSS signing	SHA-1 / 256 / 384 / 512 with caller salt length	All providers	B	Covered	—	—
RSA-08	PSS with SHA-224	SHA-224 salt and MGF1	Unknown	C	Covered	—	—
RSA-09	PKCS#1 v1.5 decryption	AT_KEYEXCHANGE keys	All providers	A	Covered	—	—
RSA-10	OAEP decryption	SHA-1 / 224 / 256 / 384 / 512	All providers	A	Covered	—	—
RSA-11	OAEP application label	pbLabel / cbLabel passed through	Unknown	C	Covered	—	—
RSA-12	Raw RSA (no padding)	CKM_RSA_X_509	Some HSMs	C	Not covered	SoftHSM2 does not implement CKM_RSA_X_509. Blocked by the backend not by the KSP; would work against an HSM that offers it. Since phase 4 this is no longer a reason not to implement it: the capability probe means the mechanism can be wired and gated on CKM_RSA_X_509 being advertised, staying dark on SoftHSM2 and lighting up on a token that has it — exactly how ML-DSA is handled.	M
ASYMMETRIC ECDSA							
ECDSA-01	ECDSA P-256	secp256r1; 64-byte signature	All providers	A	Covered	—	—
ECDSA-02	ECDSA P-384	secp384r1; 96-byte signature	All providers	A	Covered	—	—
ECDSA-03	ECDSA P-521	secp521r1; 132-byte signature	All providers	A	Covered	—	—
ECDSA-04	Brainpool curves	brainpoolP256r1 / 384r1 / 512r1	Entrust; Utimaco (device)	C	Covered	—	—
ECDSA-05	secp256k1	Bitcoin curve	Entrust (device)	C	Covered	—	—
ECDSA-06	DER to r s conversion	PKCS#11 returns DER; CNG wants raw	All providers	A	Covered	—	—
KEY AGREEMENT							
ECDH-01	ECDH P-256 / P-384 / P-521	CKM_ECDH1_DERIVE with CKD_NULL	MS Software KSP; Utimaco	B	Covered	—	—
ECDH-02	SecretAgreement / DeriveKey / FreeSecret	NCRYPT_█SECRET_█HANDL E lifecycle	MS Software KSP; Utimaco	B	Covered	—	—
ECDH-03	BCRYPT_KDF_RAW_SECRET	Raw Z value returned	MS Software KSP	B	Covered	—	—

ID	Feature	Detail	Market support	Ev	Status	Gap — what closing it would take	Eff
ECDH-04	BCRYPT_KDF_HASH	SP 800-56A concatenation KDF	MS Software KSP	B	Covered	—	—
ECDH-05	BCRYPT_KDF_HMAC / TLS_PRF / HKDF	Protocol-specific KDFs	MS Software KSP	B	Covered	—	—
ECDH-06	Finite-field Diffie-Hellman	DH 512-4096	MS Software KSP	B	Not covered	Out of scope by choice. ECDH supersedes it and no surveyed HSM KSP exposes it. Would need CKM_██DH_██PKCS_██KEY_██PAIR_██GEN plus CKM_██DH_██PKCS_██DERIVE.	L
ASYMMETRIC EDDSA							
EDDSA-01	Ed25519 signing	64-byte raw signature	None	A	Partial	Works against SoftHSM2 but uses the non-standard identifier EDDSA_ED25519. CNG defines no EdDSA algorithm ID so no stock application can request it. Nothing to fix in the KSP; blocked on Microsoft.	N/A
EDDSA-02	Ed448 signing	114-byte raw signature	None	A	Partial	Same constraint as EDDSA-01.	N/A
EDDSA-03	X25519 key agreement	Curve25519 ECDH	MS CNG has BCRYPT_EC C_CURVE_25519	B	Covered	—	—
ASYMMETRIC DSA							
DSA-01	DSA signing	512-1024 bits	MS Software KSP	B	Not covered	Deliberately out of scope. Deprecated in modern Windows PKI and absent from every HSM KSP surveyed.	L
POST-QUANTUM							
PQC-01	ML-DSA (Dilithium)	FIPS 204	Microsoft Software KSP (GA late 2025; AD CS ML-DSA enrolment May 2026). Thales Luna firmware 7.9+ and Entrust nShield XC/5/nSaaS ship it in the device; whether their CNG providers expose it is not established	B	Partial	ML-DSA is wired through the KSP and gated on the capability probe: mechanism resolution, CKA_PARAMETER_SET key generation for all three parameter sets, and signing. Unreachable on SoftHSM2 2.7.0, which defines the mechanisms and implements none of them; reachable on a PKCS#11 v3.2 token with no rebuild. Partial because public key export is refused — the CNG post-quantum key blob layout is in a Windows SDK header not available here — and because no PQC token was available to test against.	M
PQC-02	ML-KEM (Kyber)	FIPS 203	Microsoft CNG GA (BCrypt side). Thales and Entrust device firmware. No KSP-side exposure established for any provider	B	Not covered	NOT the same blocker as PQC-01. The probe recognises CKM_ML_KEM, so the backend is not what stops this: encapsulation and decapsulation have no slot in the NCRYPT_██KEY_██STORAGE_██FUNCTION_██TABLE this provider implements, so a KSP can hold an ML-KEM key with no entry point to use it. Whether a newer ncrypt_provider.h adds such slots cannot be determined here — same missing header as BUILD-01.	L
PQC-03	LMS / XMSS	Stateful hash-based signatures	Thales Luna, Entrust nShield (device claims only)	C	Not covered	Blocked at the backend, unlike PQC-01: no PKCS#11 v3.2 token surveyed implements LMS or XMSS. Stateful schemes also need one-time-key state persistence the KSP has no model for, and a lost state update forges signatures.	L
SYMMETRIC AES							
AES-01	AES key generation	128 / 192 / 256 bits	None via KSP	A	Covered	—	—
AES-02	AES-ECB	No IV	None via KSP	A	Covered	—	—
AES-03	AES-CBC	16-byte IV	None via KSP	A	Covered	—	—
AES-04	AES-CBC with padding	NCRYPT_██PAD_██CIPHER_██F LAG	None via KSP	A	Covered	—	—
AES-05	AES-GCM	12-byte nonce; 128-bit tag; AAD	None via KSP	A	Covered	—	—
AES-06	AES-CTR	16-byte counter block	None via KSP	A	Partial	Works but ChainingModeCTR is a KSP extension. CNG defines no standard counter-mode string so no stock application will select it.	N/A

ID	Feature	Detail	Market support	Ev	Status	Gap — what closing it would take	Eff
AES-07	AES-CCM	Authenticated mode	None via KSP	A	Not covered	Returns NTE_NOT_SUPPORTED. SoftHSM2 2.7.0 does not implement CKM_AES_CCM so this is a backend blocker not a KSP one. Since phase 4 this is no longer a reason not to implement it: the capability probe means the mechanism can be wired and gated on CKM_AES_CCM being advertised, staying dark on SoftHSM2 and lighting up on a token that has it — exactly how ML-DSA is handled.	M
AES-08	AES-CFB	Cipher feedback	None via KSP	A	Not covered	Returns NTE_NOT_SUPPORTED. No SoftHSM2 mechanism; would need CKM_AES_CFB128 which 2.7.0 lacks. Since phase 4 this is no longer a reason not to implement it: the capability probe means the mechanism can be wired and gated on CKM_AES_CFB128 being advertised, staying dark on SoftHSM2 and lighting up on a token that has it — exactly how ML-DSA is handled.	M
AES-09	AES key wrap	RFC 3394 / 5649	Some HSMS (device)	A	Partial	Implemented both ways through BCRYPT_■AES_■WRAP_■KEY_■BLOB, gated on CKM_AES_KEY_WRAP. Unwrap works on any token: a key arrives wrapped under a KEK already on the token and is created sensitive and non-extractable, so wrapping is a way in and not a way back out. Wrap is Partial by consequence, not omission — C_WrapKey answers CKR_■KEY_■UNEXTRACTABLE for every key this provider generates, because they are all CKA_■EXTRACTABLE=FALSE. It is useful only for a key the provider opened rather than created, on a token provisioned with extractable keys.	N/A
AES-10	AES-CMAC	Block-cipher MAC	Some HSMS (device)	A	Covered	—	—
SYMMETRIC MAC							
HMAC-01	HMAC-SHA1 / 256 / 384 / 512	Generic secret keys	None via KSP	A	Partial	Works against SoftHSM2 but HMAC_SHA* are our own identifiers. CNG does HMAC through BCrypt algorithm handles not KSP key algorithms.	N/A
HMAC-02	HMAC-SHA224	SHA-224 variant	None	C	Covered	—	—
SYMMETRIC LEGACY							
3DES-01	3DES / DES	56 / 112 / 168 bits	None via KSP	B	Not covered	Deliberately out of scope. Deprecated and excluded from modern HLK requirements despite SoftHSM2 supporting 14 DES mechanisms.	L
KEY FORMATS							
FMT-01	BCRYPT_RSAPUBLIC_BLOB export	Header + exponent + modulus	All providers	A	Covered	—	—
FMT-02	BCRYPT_ECCPUBLIC_BLOB export	Header + X + Y	All providers	A	Covered	—	—
FMT-03	EdDSA public blob export	Generic ECC magic + raw point	None	A	Covered	—	—
FMT-04	BCRYPT_RSAPUBLIC_BLOB import	Creates a session public key object	Vendor-specific	C	Covered	—	—
FMT-05	BCRYPT_ECCPUBLIC_BLOB import	Creates a session public key object	Vendor-specific	C	Covered	—	—
FMT-06	Private key export	RSAPUBLICPRIVATE / ECCPRIVATE	None (all refuse)	A	Not covered	Refused by design with NTE_NOT_SUPPORTED. Keys are CKA_■EXTRACTABLE=FALSE. This is correct HSM behaviour not a gap.	N/A
FMT-07	Private key import	Wrapping an external private key onto the token	Some HSMS	C	Not covered	Returns NTE_NOT_SUPPORTED. Would need C_CreateObject with private material or C_UnwrapKey. Contradicts the HSM posture; only add if a migration workflow demands it.	L
FMT-08	BCRYPT_KEY_DATA_BLOB	Raw symmetric key material	MS Software KSP	B	Partial	Import is implemented: BCRYPT_■KEY_■DATA_■BLOB creates a CKO_SECRET_KEY via C_CreateObject with CKA_VALUE, AES for 16/24/32-byte material and a generic secret otherwise. Export is refused and will stay refused — every key this provider creates or imports is CKA_■EXTRACTABLE=FALSE, which is the point of the HSM model. Making export work would mean creating extractable keys, weakening every deployment to satisfy one blob type.	N/A
FMT-09	PKCS#8 blob	NCRYPT_■PKCS8_■PRIVATE_■KEY_■BLOB	MS Software KSP	B	Not covered	Same constraint as FMT-07: it is a private-key format and keys are non-extractable.	L
FMT-10	OPAQUETRANSFERT blob	Vendor-opaque key transport	Vendor-specific	C	Not covered	Would let keys move between instances of the same provider. Needs a defined wrapping format; no standard to follow.	L

ID	Feature	Detail	Market support	Ev	Status	Gap — what closing it would take	Eff
KEY PROPERTIES							
PROP-01	NCRYPT_ALGORITHM_PROPERTY	Read	All providers	A	Covered	—	—
PROP-02	NCRYPT_LENGTH_PROPERTY	Read and write before finalize	All providers	A	Covered	—	—
PROP-03	NCRYPT_KEY_TYPE_PROPERTY	AT_SIGNATURE / AT_KEYEXCHANGE	All providers	A	Covered	—	—
PROP-04	NCRYPT_NAME / UNIQUE_NAME	Key label	All providers	A	Covered	—	—
PROP-05	NCRYPT_EXPORT_POLICY_PROPERTY	Always 0 (non-exportable)	All providers	A	Covered	—	—
PROP-06	NCRYPT_KEY_USAGE_PROPERTY	Signing / decrypt / key agreement flags	All providers	A	Covered	—	—
PROP-07	NCRYPT_ALGORITHM_GROUP_PROPERTY	RSA / ECDSA / ECDH / EDDSA / AES / HMAC	All providers	A	Covered	—	—
PROP-08	NCRYPT_CHAINING_MODE_PROPERTY	Read and write on symmetric keys	MS Software KSP (BCrypt)	A	Covered	—	—
PROP-09	NCRYPT_INITIALIZATION_VECTOR	Read and write on symmetric keys	MS Software KSP (BCrypt)	A	Covered	—	—
PROP-10	NCRYPT_BLOCK_LENGTH_PROPERTY	AES block size	MS Software KSP	B	Covered	—	—
PROP-11	NCRYPT_SECURITY_DESCRIPTOR_PROPERTY	Per-key ACL	MS Software KSP	B	Not covered	Returns NTE_NOT_SUPPORTED. PKCS#11 has no ACL model; would need a side-channel store mapping key labels to SDDL strings.	L
PROP-12	NCRYPT_UI_POLICY_PROPERTY	Strong-key protection prompt	MS Software KSP; smart cards	B	Not covered	Depends on PromptUser (IFACE-07). Same UI-thread requirement.	L
PROP-13	NCRYPT_PIN_PROPERTY	Per-key PIN	Smart card KSPs	A	Covered	—	—
PROP-14	NCRYPT_CERTIFICATE_PROPERTY	Certificate stored alongside the key	Smart card KSPs	A	Covered	—	—
PROP-15	NCRYPT_WINDOW_HANDLE_PROPERTY	Parent HWND for dialogs	Smart card KSPs	B	Not covered	Only meaningful with PromptUser (IFACE-07).	L
PROP-16	NCRYPT_USE_COUNT_PROPERTY	Key usage counter	Some HSMS	C	Not covered	Would need a counter object on the token incremented per operation. SoftHSM2 has no atomic counter primitive.	L
KEY LIFECYCLE							
LIFE-01	CreatePersistedKey	Immediate and deferred generation	All providers	A	Covered	—	—
LIFE-02	FinalizeKey	Deferred generation trigger	All providers	A	Covered	—	—
LIFE-03	OpenKey by name	Asymmetric and symmetric	All providers	A	Covered	—	—
LIFE-04	DeleteKey	Destroys all token objects	All providers	A	Covered	—	—
LIFE-05	EnumKeys	Paged enumeration with state	All providers	A	Covered	—	—
LIFE-06	EnumAlgorithms	NCryptEnumAlgorithms support	All providers	B	Covered	—	—
LIFE-07	IsAlgSupported	NCryptIsAlgSupported	All providers	B	Covered	—	—

ID	Feature	Detail	Market support	Ev	Status	Gap — what closing it would take	Eff
LIFE-08	Machine vs user key scope	NCRYPT_■MACHINE_■KEY_■FLAG	All providers	B	Partial	NCRYPT_■MACHINE_■KEY_■FLAG is honoured on CreatePersistedKey, OpenKey and EnumKeys: the scope becomes a CKA_LABEL prefix ("m/name" or "u/name"), so a machine key and a user key of the same name are distinct objects instead of silently aliasing. Enumeration filters by scope and reports unprefix names. This is NAMESPACE, NOT ISOLATION — anyone who can log into the token can read either scope. Real isolation needs an ACL model PKCS#11 does not provide within a slot, and no amount of label work supplies one.	L
OPERATIONAL							
OPS-01	Hardware key protection	Tamper-resistant storage	All commercial providers	A	Not covered	Fundamental: SoftHSM2 is a software token using encrypted SQLite. Only fixable by pointing the KSP at a real HSM which the PKCS#11 abstraction already allows.	N/A
OPS-02	FIPS 140-2/3 validation	Certified cryptographic module	All commercial providers	B	Not covered	Follows from OPS-01. Requires certified hardware and a formal validation programme.	N/A
OPS-03	Authenticode-signed binary	Commercial code-signing certificate	All commercial providers	A	Not covered	Signing is fully toolled: tools/sign_ksp.ps1 signs, timestamps and verifies, taking a certificate from the Windows store, a PFX (password from the environment, never a parameter) or Azure Trusted Signing, and the CI windows job signs automatically once the KSP_■SIGN_■PFX_■BASE64 secret exists. What remains is not engineering: an OV or EV code-signing certificate must be bought by a verified legal entity. Nothing in this repository can supply that, so the row stays Not covered — the binary is unsigned.	N/A
OPS-04	Multiple slots / tokens	Selecting among several tokens	All commercial providers	B	Covered	—	—
OPS-05	Key attestation	TPM / HSM-signed key origin proof	MS Platform Crypto; some HSMs	B	Not covered	No PKCS#11 standard mechanism exists. Vendor-specific and unreachable through a portable PKCS#11 layer.	L
OPS-06	Load balancing / HA	Multiple HSM members	Thales; Entrust	C	Not covered	Handled inside vendor PKCS#11 libraries. Our KSP would inherit it automatically when pointed at such a library; nothing to implement.	N/A
OPS-07	Session pool concurrency	Concurrent NCrypt calls	All providers	A	Covered	—	—
OPS-08	Re-initialisation without restart	Recover from a bad DLL path	Vendor-specific	A	Covered	—	—
OPS-09	PIN caching / re-login	Session recovery after token logout	All commercial providers	C	Covered	—	—
OPS-10	Logging / diagnostics	Trace output for support	All providers	A	Covered	—	—
OPS-13	Runtime capability discovery	Advertised algorithms reflect what the token implements, not a compiled-in list	All commercial providers (implicit — each ships for its own device)	A	Covered	—	—
OPS-14	Backend portability	Any PKCS#11 v2.40+ module can back the provider (KSP_PKCS11_LIB)	Not offered — commercial providers are bound to their own hardware	A	Covered	—	—

Maintaining this document

The source of truth is `softhsm_ksp/docs/feature-matrix.csv`. One row per capability. To record that a gap has been closed, change that row's *status* to Covered and clear its *gap_solution*; to add a newly observed market feature, append a row with a fresh ID. Then regenerate:

```
python3 generate_feature_matrix.py
```

Every number on page 1 is computed from the CSV at generation time, so the summary cannot fall out of step with the table. Because the CSV is plain text, a change to coverage shows up as a one-line diff in review rather than an opaque binary change.

Actionable gaps by effort: **2 small, 6 medium, 17 large**. Rows marked N/A are excluded — they are deliberate positions (private key export is refused by design) or external blockers (CNG defines no EdDSA identifier; SoftHSM2 is a software token).

A caution on the market columns. An HSM datasheet is not a statement about its CNG provider. AWS CloudHSM performs AES, DES3 and HMAC in hardware, yet its KSP answers NCryptIsAlgSupported for exactly four identifiers. Every grade-C cell here carries that uncertainty. See docs/11-market-comparison.md for the full audit and its sources.